

Assignment/Homework: Responsible Data Use Review

Goal

This week, your team will examine your capstone project through the lens of **ethics, privacy, security, bias, and responsible data use**. The purpose is to identify risks early and propose concrete design or governance decisions that address them.

Team Task

Review your current project and prepare a short assessment covering the following areas.

1. Data and privacy

- What data are you collecting, storing, or processing?
- Does the project involve personal, sensitive, or potentially identifying data?
- Are you collecting more data than you actually need?
- What are your plans for access control, retention, and deletion?

2. Security

- What are the main security risks in your architecture?
- Where are credentials, API keys, or secrets stored?
- Who should have access to raw data?
- What security controls should be added to your pipeline, database, API, or application?

3. Bias and fairness

- Could the data underrepresent or misrepresent particular groups or situations?
- What sampling, measurement, historical, or labeling biases might exist?
- Could your project's outputs have different consequences for different users or groups?

4. Ethics and responsible use

- Who benefits from the system?
- Who could be harmed if the system produces incorrect or misleading results?
- Are there uses of your system that should be discouraged or explicitly prohibited?
- What assumptions should users know about before relying on your results?

Use at Least Two External References

Consult at least **two** of the following resources and explain briefly how they apply to your project:

- NIST AI Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>
- NIST Privacy Framework: <https://www.nist.gov/privacy-framework>
- ACM Code of Ethics: <https://www.acm.org/code-of-ethics>
- OWASP Top 10: <https://owasp.org/www-project-top-ten/>
- OECD AI Principles: <https://oecd.ai/en/ai-principles>
- GDPR principles: <https://eur-lex.europa.eu/eli/reg/2016/679/oj>
- FAIR Data Principles: <https://www.go-fair.org/fair-principles/>
- Datasheets for Datasets: <https://arxiv.org/abs/1803.09010>
- Model Cards for Model Reporting: <https://arxiv.org/abs/1810.03993>

Deliverable for Next Friday’s Lab

Prepare a **3–5 minute team presentation** with approximately **3–4 slides**:

1. **Project context:** briefly describe your project and the data it uses.
2. **Top risks:** identify the 2–4 most important ethical, privacy, security, or bias concerns.
3. **Responsible design decisions:** describe specific actions your team will take to reduce these risks.
4. **Open question:** end with one issue where your team would like feedback from the class.

What Makes a Strong Submission

A strong presentation should move beyond statements such as “we will protect privacy” or “we will avoid bias.” Connect each concern to a specific part of your project and propose a concrete technical, organizational, or documentation response.

Guiding question: What could go wrong with our project, and what are we doing about it?